

RECAP & BEST PRACTICES

Tengku Muhammad Hanis Bin Tengku Mokhtar, PhD

November 12, 2026

Content

- Recap
- Best practices in R

Recap

- RStudio, Posit Cloud, and Google Colab — tools you can use
- We clean and reshape data — selecting columns, filtering rows, and creating new variables
- We summarise data — descriptive statistics
- We visualise data — histograms, bar plots, box plots, and scatter plots
- We combine text, code, and results in one document (Quarto)
- We saw how AI can help write and fix R code — as a helper, not a replacement
- Do not be 100% dependant on AI — AI may take your creativity and critical thinking away

Best practices in R

1. One project, one folder
2. Raw data is read-only — never touch the original CSV, all cleaning happens in code
3. Clean before you compute — check duplicates (need a unique ID), missingness (anyNA, miss_var_summary), and variable types (str) first
4. Script everything — every table, figure, and p-value must be regenerable by re-running the script; if it can't be rerun, it doesn't exist
5. Comment as you go — # headers per step make scripts readable for your supervisor and future-you (your own scripts model this well)

6. Use <- and consistent naming — lowercase with underscores
7. Plot before you model — distributions and outliers change which statistics are appropriate
8. Choose the right summary — mean/SD for symmetric data, median/IQR for skewed; report them, don't guess
9. Verify AI-generated code — run it, check output, confirm the function exists

Thank You !